

DD_SUB-WF-RESUME-01 - Subscription Resume

Developer implementation guide: DD_SUB-WF-RESUME-01-DEV_IMPL_GUIDE-v1.2.md.

1. Purpose

SUB-WF-RESUME-01 defines how a suspended subscription is restored to active service.

Resume moves a subscription from:

SUSPENDED -> ACTIVE

The technical saga uses a transient status:

SUSPENDED -> PENDING_RESUME -> ACTIVE

Use this operation when:

- A customer asks to resume a voluntary pause.
- A call-center user resumes a customer pause.
- A BO admin force-resumes any suspension.
- BIL-04 resumes after non-payment debt is cleared.
- The scheduler auto-resumes a fixed-duration or scheduled pause.

2. What Resume Closes

Resume closes an open row in subscription_pause_history.

An open row is:

actual_resume_at IS NULL

Resume updates the row with:

- actual_resume_at
- resume_reason_code
- resume_actor_user_id
- resume_actor_role
- reconnection_fee_billable_event_id, when a reconnection fee applies
- admin-force audit fields, when ADMIN_FORCE_RESUME is used

The open row tells resume why the subscription is suspended. That value is called origin_intent.

3. Ownership

This DD owns:

- Resume trigger API: POST /api/subscriptions/{subscription_id}/resume.
- Resume trigger taxonomy.
- Origin-intent gating rules.
- subscription_resume_config table.
- Pause-history close-out contract.
- Resume business rules.
- Resume domain events.
- Resume worker usage.

This DD does not own:

- WIK-specific BPMN. See DD_SUB-WF-RESUME-01-FLOW-WIK.

- Subscription master state. See SUB-LM-01.
- Operation orchestration framework. See SUB-WF-FRAMEWORK-01.
- Worker topic contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Pause history table creation. See SUB-WF-PAUSE-01 / shared lifecycle persistence.
- Non-payment suspension creation. See SUB-WF-SUSPEND-NP-01.
- Network reactivation implementation. See FUL-03.
- Billing debt-cleared decision. See BIL-04.
- Billing fee definitions. See BIL-CFG-01.
- Notification templates and channel delivery. See NOT-01.
- Scheduled scan ownership. See foundation scheduler.

4. Deployment Unit Clarification

Resume is not a standalone service.

It is deployed as part of the subscription workflow capability:

- Resume route/controller in SUB-WF API.
- Resume config migration.
- Resume BPMN, such as sub-resume-wik.
- Resume Drools package, such as rules.subscription.resume.wik.
- Shared worker apps from the worker catalog.

The same worker app can execute multiple resume, pause, and activation topics. Keep metrics, logs, retries, and alerts visible by topic.

5. Resume Triggers

Trigger	Meaning	Allowed roles
CUSTOMER_REQUESTED_RESUME	Customer or support resumes a voluntary customer pause.	CUSTOMER_SELF_SERVICE, CALL_CENTER, SUBSCRIPTION_OPERATOR
ADMIN_FORCE_RESUME	Admin/operator override resumes any supported suspension type.	SUBSCRIPTION_ADMIN, SUBSCRIPTION_OPERATOR
DUNNING_RESUME	Billing resumes a non-payment suspension after debt is cleared.	BILLING_INTERNAL
SCHEDULED_AUTO_RESUME	Scheduler resumes a pause when resume_scheduled_at arrives.	SCHEDULER_INTERNAL

CUSTOMER_SELF_SERVICE also requires:

- subscription_resume_config.customer_self_service_enabled = true
- open pause row has origin_intent = CUSTOMER_REQUESTED_PAUSE

6. Origin-Intent Gating

Resume must read the open subscription_pause_history row before starting the BPMN.

origin_intent	Meaning
CUSTOMER_REQUESTED_PAUSE	Voluntary customer pause created by PAUSE.
ADMIN_PAUSE	Admin pause created by PAUSE.
SUSPEND_NP	Non-payment suspension created by SUSPEND-NP.

Allowed matrix:

Resume trigger	CUSTOMER_REQUESTED_PAUSE	ADMIN_PAUSE	SUSPEND_NP
CUSTOMER_REQUESTED_RESUME	Allowed	Blocked: ADMIN_PAUSE_BLOCKS_CUSTOMER_RESUME	Blocked: SYSTEM_MANAGED_SUSPENSION
ADMIN_FORCE_RESUME	Allowed	Allowed	Allowed
DUNNING_RESUME	Blocked: NOT_A_DUNNING_SUSPENSION	Blocked: NOT_A_DUNNING_SUSPENSION	Allowed
SCHEDULED_AUTO_RESUME	Allowed only when resume_scheduled_at is set	Allowed only when resume_scheduled_at is set	Blocked: NOT_A_SCHEDULED_RESUME

7. Data Model

7.1 subscription_resume_config

What this table is for:

subscription_resume_config stores operator-level resume behavior. It controls customer self-service, whether customer resumes require a reconnection fee, whether dunning resume re-checks debt with billing, and whether success notification is enabled.

Use this table when:

- Validating CUSTOMER_SELF_SERVICE resume.
- Deciding whether CUSTOMER_REQUESTED_RESUME may charge reconnection fee.
- Deciding whether DUNNING_RESUME should ask BIL-01 to re-check debt.
- Deciding whether BPMN should call notification-send.

Do not use this table for:

- Per-customer debt balance.
- Current subscription state.
- Pause reason eligibility.
- Network provisioning decisions.

```
CREATE TABLE subscription_resume_config (
  operator_code          TEXT PRIMARY KEY,
  customer_self_service_enabled  BOOLEAN NOT NULL DEFAULT FALSE,
  reconnection_fee_on_customer_resume  BOOLEAN NOT NULL DEFAULT FALSE,
  dunning_resume_verify_debt_cleared  BOOLEAN NOT NULL DEFAULT TRUE,
  customer_notification_enabled  BOOLEAN NOT NULL DEFAULT TRUE,
  updated_at             TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_by             TEXT NOT NULL
);
```

Column descriptions:

Column	Purpose
operator_code	Operator this config applies to.
customer_self_service_enabled	Allows customer portal/app to start CUSTOMER_REQUESTED_RESUME.
reconnection_fee_on_customer_resume	Enables pay-first reconnection fee for customer-triggered resume.
dunning_resume_verify_debt_cleared	Adds BIL-01 debt-cleared verification for DUNNING_RESUME.
customer_notification_enabled	Enables customer notification after successful resume.
updated_at	Last update timestamp.
updated_by	User/service that last updated the row.

Sample data:

operator_code	customer_self_service_enabled	reconnection_fee_on_customer_resume	dunning_resume_verify_debt_cleared	customer_notification_enabled
WIK	true	false	true	true
WUG	false	false	true	true
WTZ	false	false	true	true

operator_code	customer_self_service_enabled	reconnection_fee_on_customer_resume	dunning_resume_verify_debt_cleared	customer_notification_enabled
YASSN	false	false	true	true

Sample meaning:

- WIK allows self-service resume.
- WIK does not charge reconnection fee by default in v1.2.
- WIK still asks billing to verify dunning debt is cleared.
- Other operators are conservative placeholders until their flow satellites are defined.

7.2 subscription_pause_history Close-Out Fields

What this data is for:

Resume uses the open pause-history row as the business source for why the subscription is suspended. Resume then closes that row after network reactivation and state commit succeed.

Use this data when:

- Validating origin-intent gating.
- Showing support why a resume is blocked.
- Closing the suspension period.
- Auditing admin force resume.
- Letting BIL-02 compute cycle accrual after resume.

Resume reads these fields before starting:

Column	Purpose	Sample
pause_id	Row to close.	pause_01HZ_KAMAU_SUSPNP
subscription_id	Suspended subscription.	sub_01HZ_KAMAU_FIBER
origin_intent	Why suspension exists.	SUSPEND_NP
system_managed	Whether suspension was system-managed.	true
resume_scheduled_at	Scheduler resume time, if any.	2026-04-15T10:00:00Z
suspended_at	When suspension started.	2026-06-15T02:00:18Z
actual_resume_at	Must be NULL before resume.	NULL
dunning_cycle_reference	Billing dunning reference, if non-payment suspension.	bil04-dunning-wave-2026-06-15-L3
outstanding_debt_amount	Debt snapshot at suspension time.	3500.00

Resume writes these fields after success:

Column	Purpose	Sample
actual_resume_at	Actual resume timestamp.	2026-06-22T10:00:18Z
resume_reason_code	Resume reason supplied by caller or system.	DEBT_CLEARED_VIA_MPESA
resume_actor_user_id	User/service that resumed.	bil04-svc-account
resume_actor_role	Role that resumed.	BILLING_INTERNAL
reconnection_fee_billable_event_id	BIL event for reconnection fee, if any.	bill_evt_01HZ_RECONN
admin_force_resume	Whether admin override was used.	false
admin_force_resume_reason	Admin reason, if forced.	Customer emergency override
dunning_cycle_reference_at_resume	Dunning reference that triggered resume.	bil04-resume-2026-06-22

Example before resume:

pause_id	subscription_id	origin_intent	suspended_at	actual_resume_at	resume_scheduled_at
pause_01HZ_KAMAU_SUSPNP	sub_01HZ_KAMAU_FIBER	SUSPEND_NP	2026-06-15T02:00:18Z	NULL	NULL

Example after resume:

pause_id	subscription_id	origin_intent	actual_resume_at	resume_reason_code	resume_actor_role
pause_01HZ_KAMAU_S USPNP	sub_01HZ_KAMAU_FIB ER	SUSPEND_NP	2026-06-22T10:00:1 8Z	DEBT_CLEARED_VIA_M PESA	BILLING_INTERNAL

8. Business Rules

8.1 State Rules

Rule	Description
R-RESUME-S-1	Resume requires subscription status SUSPENDED. Other statuses return 409 INVALID_STATE_FOR_RESUME.
R-RESUME-S-2	Transition is SUSPENDED -> PENDING_RESUME -> ACTIVE.
R-RESUME-S-3	Resume is mutually exclusive with other state-changing operations on the same subscription.
R-RESUME-S-4	During PENDING_RESUME, current_transition_type = RESUME and current_transition_reason_code = resumeTrigger.

8.2 Trigger Rules

Rule	Description
R-RESUME-T-1	Trigger must be one of the four values in section 5.
R-RESUME-T-2	Actor role must match the trigger role set.
R-RESUME-T-3	CUSTOMER_SELF_SERVICE requires operator config enabled.
R-RESUME-T-4	SCHEDULED_AUTO_RESUME requires an open row with resume_scheduled_at IS NOT NULL.

8.3 Origin-Intent Rules

Rule	Description
R-RESUME-OI-1	Resume must read the open pause-history row before starting BPMN.
R-RESUME-OI-2	No open row returns 409 NO_OPEN_PAUSE_HISTORY.
R-RESUME-OI-3	Trigger and origin_intent must match the matrix in section 6.
R-RESUME-OI-4	ADMIN_FORCE_RESUME on SUSPEND_NP preserves origin_intent = SUSPEND_NP and writes admin-force audit fields.

8.4 Billing Rules

Rule	Description
R-RESUME-B-1	BPMN sends RESUME_INTENT for customer/admin resume and RESUME_NP_INTENT for dunning resume.
R-RESUME-B-2	If reconnection_fee_on_customer_resume = true and trigger is CUSTOMER_REQUESTED_RESUME, BPMN waits for reconnection fee payment.
R-RESUME-B-3	If dunning_resume_verify_debt_cleared = true, DUNNING_RESUME asks BIL-01 to confirm debt is cleared.
R-RESUME-B-4	ADMIN_FORCE_RESUME normally has no fee, but still sends billing intent for audit and cycle handling.

8.5 Restrictions And Cycle Rules

Rule	Description
R-RESUME-RC-1	Restrictions are not automatically reapplied on resume. If needed, call RESTRICT after resume.
R-RESUME-RC-2	Cycle anchor and cycle model are unchanged by resume.
R-RESUME-RC-3	BIL-02 resumes cycle accrual from the SubscriptionResumed event.

9. REST API

9.1 Trigger Resume

```
POST /api/subscriptions/{subscription_id}/resume HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Idempotency-Key: resume-sub_01HZ_KAMAU-bil04-payment-cleared-20260622
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "resumeTrigger": "DUNNING_RESUME",
  "resumeReasonCode": "DEBT_CLEARED_VIA_MPESA",
  "dunningCycleReference": "bil04-resume-2026-06-22",
  "notes": "Customer paid INV_01HZ KES 3,500 via M-Pesa",
  "initiatingWorkflowKind": "BIL_04_DUNNING",
  "initiatingWorkflowRef": "bil04-resume-2026-06-22"
}
```

Actor user and role come from the token, not the public request body.

Success:

```
202 Accepted

{
  "operationId": "subop_01HZ_KAMAU_RESUME",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "RESUME",
  "resumeTrigger": "DUNNING_RESUME",
  "bpmnProcessKey": "sub-resume-wik",
  "currentState": "INITIATED",
  "finalState": null
}
```

Common errors:

HTTP	Code	Meaning
400	UNKNOWN_RESUME_TRIGGER	Trigger is not recognized.
400	NO_SCHEDULED_RESUME_TIME	Scheduler tried to resume an open-ended pause.
403	ROLE_NOT_AUTHORIZED_FOR_TRIGGER	Actor role cannot use this trigger.
403	ADMIN_PAUSE_BLOCKS_CUSTOMER_RESUME	Customer tried to resume admin pause.
403	SYSTEM_MANAGED_SUSPENSION	Customer tried to resume dunning suspension.
403	NOT_A_DUNNING_SUSPENSION	BIL-04 tried to resume a voluntary/admin pause.
403	NOT_A_SCHEDULED_RESUME	Scheduler tried to resume a non-scheduled row.
409	INVALID_STATE_FOR_RESUME	Subscription is not SUSPENDED.
409	NO_OPEN_PAUSE_HISTORY	Subscription has no open pause-history row.
409	OPERATION_IN_PROGRESS	Another state-changing operation is in-flight.
409	IDEMPOTENCY_KEY_REUSED	Same key was reused with different request hash.
422	DEBT_NOT_CLEARED	Dunning resume requested but billing says debt remains.

9.2 Example Customer Blocked From Dunning Resume

```
403 Forbidden

{
  "code": "SYSTEM_MANAGED_SUSPENSION",
  "openPauseOriginIntent": "SUSPEND_NP",
  "dunningCycleReference": "bil04-dunning-wave-2026-06-15-L3",
  "guidance": "This subscription is suspended for non-payment. Settle the outstanding balance to restore service."
}
```

10. Worker Usage In This Operation

The worker list is a logical BPMN task list, not a deployment list.

Worker topic	Why resume uses it
drools-decide	Validate status, trigger, role, origin-intent, fee, and debt rules.

Worker topic	Why resume uses it
sub-lm-put-pending-status	Move to PENDING_RESUME.
sub-lm-compute-cycle-window	Build billing context.
bil01-emit-intent	Ask billing to decide resume intent, fee, and debt-cleared validation.
bil01-charge-pre-authorize	Optional pay-first reconnection fee.
bil01-confirm-charge	Optional fee confirmation after payment.
bil01-revert-pending-charge	Compensation for reconnection fee timeout/failure.
ful-call-activate	Call FUL-03 to reactivate network service.
sub-lm-put-master-fields	Close pause-history row and update audit fields.
sub-lm-commit-state	Commit subscription to ACTIVE.
sub-lm-revert-to-prior	Revert to SUSPENDED during compensation.
kafka-emit	Emit resume success/rejected event.
notification-send	Notify customer/admin, reliability-isolated.
subscription-operation-finalize	Mark operation terminal.

11. Kafka Events

11.1 SubscriptionResumed

Topic:

sophix.subscription.subscription.resumed

Emitted after:

- FUL-03 reactivation succeeds.
- Pause-history row is closed.
- SUB-LM commits status to ACTIVE.

Sample:

```
{
  "eventType": "SubscriptionResumed",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "timestamp": "2026-06-22T10:00:18Z",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "operationId": "subop_01HZ_KAMAU_RESUME",
    "priorStatus": "SUSPENDED",
    "newStatus": "ACTIVE",
    "resumedAt": "2026-06-22T10:00:18Z",
    "resumeTrigger": "DUNNING_RESUME",
    "resumeReasonCode": "DEBT_CLEARED_VIA_MPESA",
    "pauseHistoryRowId": "pause_01HZ_KAMAU_SUSPNP",
    "closedPauseOriginIntent": "SUSPEND_NP",
    "pauseDurationDays": 7,
    "pauseStartedAt": "2026-06-15T02:00:18Z",
    "reconnectionFee": null,
    "adminForceResume": false,
    "initiatingActorUserId": "bil04-svc-account",
    "initiatingActorRole": "BILLING_INTERNAL"
  }
}
```

11.2 SubscriptionResumeRejected

Topic:

sophix.subscription.subscription.resumerejected

Emitted after validation, billing, payment, or fulfillment failure.

Sample:

```
{
  "eventType": "SubscriptionResumeRejected",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
```

```

    "timestamp": "2026-06-22T10:00:18Z",
    "payload": {
      "operatorCode": "WIK",
      "subscriptionId": "sub_01HZ_KAMAU_FIBER",
      "operationId": "subop_01HZ_KAMAU_RESUME",
      "resumeTrigger": "DUNNING_RESUME",
      "rejectionCode": "DEBT_NOT_CLEARED",
      "openPauseOriginIntent": "SUSPEND_NP"
    }
  }
}

```

12. Drools Package

Package pattern:

```
rules.subscription.resume.<operator>
```

For WIK:

```
rules.subscription.resume.wik
```

Required decisions:

Decision	Purpose
validate-preconditions	Status suspended, open pause row exists, no in-flight operation.
validate-trigger-role	Trigger and actor role are compatible.
validate-origin-intent	Trigger is allowed for the open pause row's origin_intent.
compute-resume-billing-intent	Chooses RESUME_INTENT or RESUME_NP_INTENT and fee/debt checks.

13. Auth

Rules:

- Resolve actor from token, not request body.
- Customer self-service requires operator config enabled.
- Dunning resume requires a BIL-04 service account.
- Scheduled auto-resume requires a scheduler service account.
- Admin force resume requires admin/operator role and a reason note.

14. Infrastructure Recommendations

Resume does not need separate infrastructure from SUB-WF.

Deploy:

Artifact	Where
Resume endpoint/controller	SUB-WF API pods.
Resume config migration	Application DB migration pipeline.
Resume BPMN files	Camunda deployment pipeline.
Resume Drools packages	Drools/KIE deployment pipeline.
Resume worker usage	Worker pods/apps from worker DD.

Monitor:

- Resume request count by trigger.
- Resume failure count by rejection code.
- Dunning resume DEBT_NOT_CLEARED count.
- Scheduled resumes missed or delayed.
- FUL-03 reactivation failure count.
- Operations stuck in PENDING_RESUME.

- Open pause-history rows with `resume_scheduled_at < now`.

Medium budget deployment recommendation:

- Run resume API inside existing SUB-WF API pods.
- Run resume workers inside the shared subscription worker deployment.
- Do not create a dedicated resume worker pod unless volume or isolation requires it.
- Keep Camunda, DB, Kafka, and Drools sizing from the foundation/framework recommendations.

15. AWS Deployment Recommendations

For AWS, deploy resume as part of the same subscription workflow stack:

Component	AWS recommendation
SUB-WF API	EKS deployment behind internal ALB/API gateway.
Worker apps	EKS deployment with HPA based on Camunda external task lag and CPU.
Camunda	EKS service using the approved foundation Camunda topology.
PostgreSQL	RDS PostgreSQL or Aurora PostgreSQL with Multi-AZ.
Kafka	MSK or approved Kafka platform.
Drools/KIE	EKS service or artifact-driven rule executor, same as foundation rule setup.
Secrets	AWS Secrets Manager or Parameter Store with IRSA.
Observability	CloudWatch Container Insights plus application metrics for operation states and worker topics.

AWS alarms:

- PENDING_RESUME older than configured SLA.
- Scheduled resume lag above SLA.
- SubscriptionResumeRejected spike.
- FUL-03 failure spike.
- Outbox backlog for `subscription.resumed`.

16. Cross-DD Dependencies

DD/module	Relationship
SUB-WF-FRAMEWORK-01	Operation rows, idempotency, concurrency, Camunda start.
DD_SUB-WF-FRAMEWORK-01-WORKERS	Worker topic contracts.
DD_SUB-WF-RESUME-01-FLOW-WIK	WIK BPMN implementation.
SUB-LM-01	Subscription state and active restrictions.
SUB-WF-PAUSE-01	Creates voluntary/admin pause-history rows that resume closes.
SUB-WF-SUSPEND-NP-01	Creates dunning suspension rows that dunning resume closes.
BIL-01	Resume intent, fee handling, and debt validation.
BIL-02	Consumes SubscriptionResumed to resume cycle accrual.
BIL-04	Calls DUNNING_RESUME after debt settlement.
FUL-03	Network/service reactivation.
NOT-01	Notifications.
Scheduler	Calls SCHEDULED_AUTO_RESUME for due pause rows.

17. Build Order

Detailed flow is in DD_SUB-WF-RESUME-01-DEV_IMPL_GUIDE-v1.2.md.

Recommended order:

1. Add `subscription_resume_config` migration and seed rows.

2. Ensure subscription_pause_history has resume close-out fields.
3. Add framework operation config for RESUME.
4. Implement resume API.
5. Implement Drools facts/decisions.
6. Deploy first BPMN, normally WIK.
7. Wire workers and message catches.
8. Add compensation paths.
9. Wire BIL-04 DUNNING_RESUME.
10. Wire scheduler SCHEDULED_AUTO_RESUME.

18. Test Checklist

- [] Resume rejects non-SUSPENDED subscriptions.
- [] Resume rejects missing open pause-history row.
- [] Trigger role gating works for all four triggers.
- [] Customer cannot self-resume admin pause.
- [] Customer cannot self-resume non-payment suspension.
- [] BIL-04 cannot dunning-resume voluntary pause.
- [] Scheduler cannot resume open-ended pause.
- [] Dunning resume validates debt when config requires it.
- [] Optional reconnection fee path waits for payment.
- [] Success closes pause-history row.
- [] Success commits subscription to ACTIVE.
- [] Success emits SubscriptionResumed.
- [] Failure emits SubscriptionResumeRejected.
- [] Notification failure does not roll back successful resume.